‹#›

1: Find the derivative of polynomials.

2: Find equations of tangents and normal to curves.

3: Identify increasing and decreasing functions.

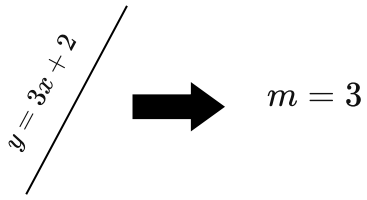
4: Find and understand the second derivative $\frac{d^2y}{dx^2}$ or $f''(x)$

5: Find stationary points and determine their nature.

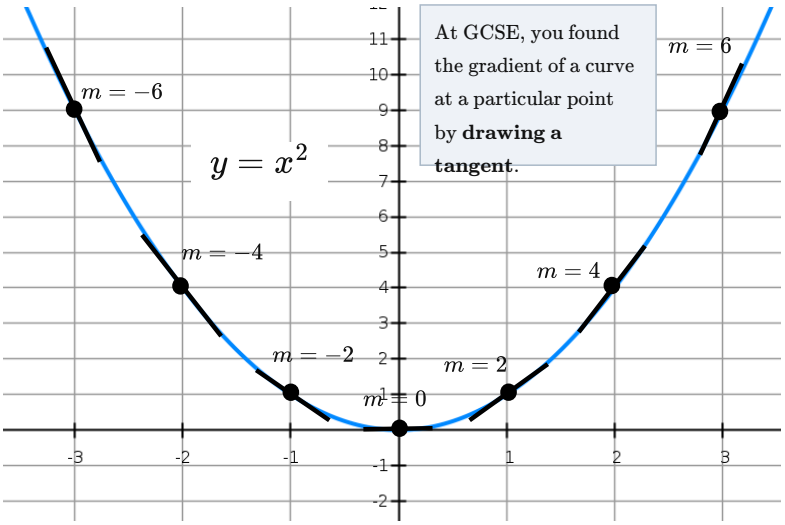
6: Sketch a gradient function.

7: Model real-life problems.

For a straight line, the gradient is **constant**:



However, for a curve **the gradient varies**. We can no longer have a single value for the gradient; **we ideally want an expression in terms of x** that gives us the gradient for any value of x (unsurprisingly known as the **gradient function**).



At GCSE, you found the gradient of a curve at a particular point by **drawing a tangent**.

x	-3	-2	-1	0	1	2	3
Gradient							

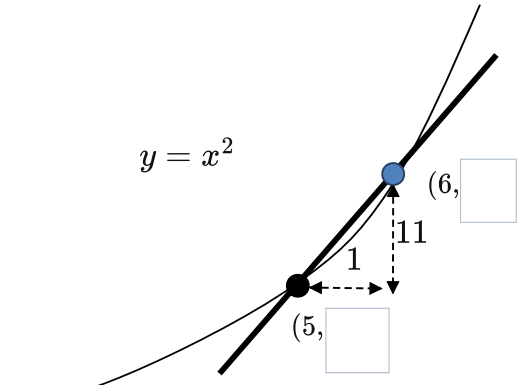
By looking at the relationship between x and the gradient at that point, can you come up with an expression, in terms of x for the gradient?

$GradientFunction =$ **2x**

Finding the Gradient Function 4

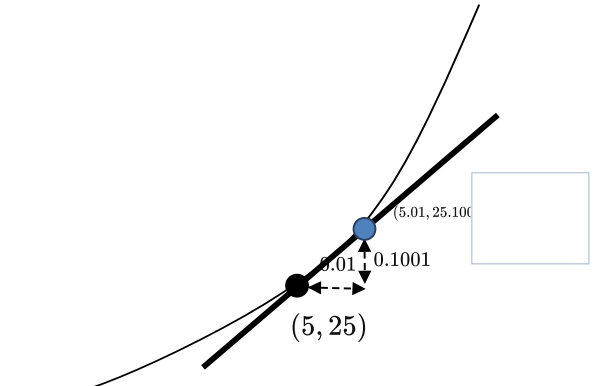
The question is then: Is there a method to work out the gradient function without having to draw lots of tangents and hoping that we can spot the rule?

To approximate the gradient on the curve $y = x^2$ when $x = 5$, we could pick a point on the curve just slightly to the right, then find the gradient between the two points:



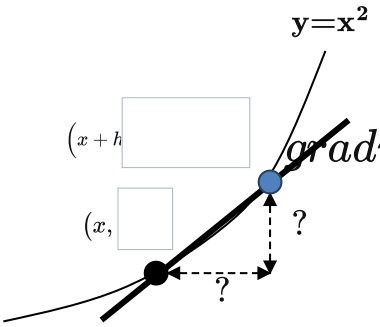
$m = \frac{\Delta y}{\Delta x} =$

As the second point gets closer and closer, the gradient becomes a better approximation of the true gradient:



The actual gradient when $x = 5$ is 10, so this approximation is damn close!

This gives us a numerical method to get the gradient **at a particular x**, but doesn't give us the gradient function in general. Let's use exactly the same method, but keep x general, and make the 'small change' (which was previously 0.01) 'h':



$$\begin{aligned} gradient &= \lim_{h\to 0} \\ &= \lim_{h} \\ &= \lim_{h} \\ &= 2x \end{aligned}$$

As always, gradient is change in y over change in x.

The lim means "*the limit of the following expression as h tends towards 0*".
For example, $\lim_{x\to\infty}\left(\frac{1}{x}\right) = 0$, because as x "tends towards" infinity, the "limiting" value of the expression is 0.

The h disappears as h tends towards 0, i.e. we can effectively treat it as 0 at this point.

And voila, we got the $2x$ we saw earlier!

✎ The gradient function, or derivative, of the curve $y = f(x)$ is written as $f\prime(x)$ or $\frac{dy}{dx}$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

The gradient function can be used to find the gradient of the curve for any value of x.

We will soon see an easier/quicker way to differentiate expressions like $y = x^3 - x$ without using 'limits'. But this method, known as **differentiating by first principles**, is now in the A Level syllabus, and you could be tested on it!

Notation Note:
Whether we use $\frac{dy}{dx}$ or $f\prime(x)$ for the gradient function depends on whether we use $y =$ or $f(x) =$ to start with:

$y = x^2 \quad \rightarrow \quad \frac{dy}{dx} = 2x$ "Leibniz's notation"
$f(x) = x^2 \quad \rightarrow \quad f'(x) = 2x$ "Lagrange's notation"

Advanced Notation Note:
Rather than h for the small change in x, the formal notation is δx. So actually:

$$f'(x) = \lim_{\delta x\to 0} \frac{f(x+\delta x)-f(x)}{\delta x}$$

So we in fact have 3 symbols for "change in"!

- Δx: any change in x (as seen in Chp5: $m = \frac{\Delta y}{\Delta x}$)
- δx: a small change in x
- dx: an infinitesimally small change in x

There's in fact a third way to indicate the gradient function, notation used by Newton: (but not used at A Level)

$$y = x^2 \quad \rightarrow \quad \dot{y} = 2x$$

So the estimated gradient using some point close by was $\frac{\delta y}{\delta x}$, but in the 'limit' as $\delta x \to 0$, $\frac{\delta y}{\delta x} \to \frac{dy}{dx}$

Example 7

The point A with coordinates $(4, 16)$ lies on the curve with equation $y = x^2$.
At point A the curve has gradient g.
a) Show that $g = \lim_{h \to 0} (8 + h)$
b) Deduce the value of g.

a

b

Test Your Understanding 8

Prove **from first principles** that the derivative of x^4 is $4x^3$.

al's

1

.

5 Prove, from first principles, that the derivative of $6x$ is 6.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.
$$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$$

6 Prove, from first principles, that the derivative of $4x^2$ is $8x$.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.
$$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$$

7 $f(x) = ax^2$, where a is a constant. Prove, from first principles, that $f'(x) = 2ax$.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.
$$f'(x) = \frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

Challenge $f(x) = \frac{1}{x}$.
a Given that $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$, show that $f'(x) = \lim_{h\to 0} \frac{-1}{x^2+xh}$.
b Deduce that $f'(x) = -\frac{1}{x^2}$.

Why couldn't we just immediately make h equal to 0 in $\lim_{h\to 0} \frac{(x+h)^2-x^2}{h}$?

If we just stick $h = 0$ in straight away:

$\lim_{h\to 0} \frac{(x+0)^2-x^2}{0} = \lim_{h\to 0} \frac{0}{0}$

Wait, uh oh…

$\frac{0}{0}$ is known as an **indeterminate form**. Whereas we know what happens with expressions like $\frac{1}{0}$ (i.e. its value tends towards infinity), indeterminate forms are bad because their values are ambiguous, and prevent lim expressions from being evaluated.

Consider $\frac{0}{0}$ for example: 0 divided by anything usually gives 0, but anything divided by 0 is usually infinity. We can see these conflict.

Can you guess some other indeterminate forms, i.e. expressions whose value is ambiguous?

Thankfully, when indeterminate forms appear in lim expressions, there are variety of techniques to turn the expression into one that doesn't have any indeterminate forms.

One simple technique, that worked in our example, is to expand and simplify. This gave us $\lim_{h\to 0} (2x + h)$ and $2x + 0$ clearly is fine. Other techniques are more advanced.
This is important when sketching harder functions: http://www.drfrostmaths.com/resources/resource.php?rid=163

Thankfully, there's a quick way to differentiate terms of the form x^n (where n is a constant) with having to use first principles every time:

! If $y = ax^n$ then $\frac{dy}{dx} = nax^{n-1}$ (where a, n are constants)
i.e. multiply by the power and reduce the power by 1

Examples:

$y = x^5 \quad \rightarrow \quad \frac{dy}{dx} =$

Power is 5, so multiply by 5 then reduce power by 5.

$f(x) = x^{\frac{1}{2}} \quad \rightarrow$

The power need not be an integer!
Remember to use $f'(x)$ not $\frac{dy}{dx}$

$y = 2x^6 \quad \rightarrow$

$f(x) = \frac{x}{x^4} =$ $\rightarrow$

$y = \sqrt{x^6} =$ $\rightarrow$

1 $y = x^7 \rightarrow \frac{dy}{dx} = 7$

2 $y = 3x^{10} \rightarrow$

3 $f(x) = \frac{x^{\frac{1}{2}}}{x^2} =$ $\rightarrow$

4 $y = ax^a \rightarrow$

5 $f(x) = \sqrt{49x^7} =$ $\rightarrow$ $f'(x) = \frac{49}{2}$

Differentiate $y = x^2 + 4x + 3$

First thing to note:
If $y = f(x) + g(x)$ then
$\frac{dy}{dx} = f'(x) + g'(x)$
i.e. **differentiate each term individually in a sum/subtraction.**

$\frac{dy}{dx} =$

$y = 4x = 4x^1$
Therefore applying the usual rule:
$\frac{dy}{dx} = 4x^0 = 4$

Alternatively, if you compare $y = 4x$ to $y = mx + c$, it's clear that the gradient is fixed and $m = 4$.

$y = 3 = 3x^0$
Therefore applying the usual rule:
$\frac{dy}{dx} = 0x^{-1} = 0$

Alternatively, if you sketch $y = 4$, the line is horizontal, so the gradient is 0.

1 $y = 2x^2 - 3x \rightarrow \frac{dy}{dx} =$

2 $y = 4 - 9x^3 \rightarrow \frac{dy}{dx} =$

3 $y = 5x + 1 \rightarrow$

4 $y = ax \rightarrow$

(where a is a constant)

5 $y = 6x - 3 + px^2 \rightarrow \frac{dy}{dx} =$

(where p is a constant)

Let $f(x) = 4x^2 - 8x + 3$

a) Find the gradient of $y = f(x)$ at the point $\left(\frac{1}{2}, 0\right)$

b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 8.

c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 4x - 5$.

a

Remember that the 'gradient function' allows you to find the gradient for a particular value of x.

b

This example is important!
Previously you used a value of x to get the gradient $f'(x)$. This time we're **doing the opposite**: using a known gradient $f'(x)$ to get the value of x. We therefore substitute $f'(x)$ for 8.

Once you have your x, you need to work out y.
Ensure you use the correct equation!

c

Let $f(x) = x^2 - 4x + 2$

a) Find the gradient of $y = f(x)$ at the point $(1, -1)$

b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 5.

c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 2 - x$.

a

b

c

5 Find the gradients of the curve $y = x^2 - 5x + 10$ at the points A and B where the curve meets the line $y = 4$.

6 Find the gradients of the curve $y = 2x^2$ at the points C and D where the curve meets the line $y = x + 3$.

7 $\mathrm{f}(x) = x^2 - 2x - 8$.

a Sketch the graph of $y = \mathrm{f}(x)$.

b On the same set of axes, sketch the graph of $y = \mathrm{f}'(x)$.

c Explain why the x-coordinate of the turning point of $y = \mathrm{f}(x)$ is the same as the x-coordinate of the point where the graph of $y = \mathrm{f}'(x)$ crosses the x-axis.

If your expression isn't a sum of x^n terms, simply manipulate it until it is!

1. Turn roots into powers:

$y = \sqrt{x} = x^{\square} \rightarrow \frac{dy}{dx} = \square$

$y = \frac{1}{\sqrt[3]{x}} = x^{\square} \rightarrow \frac{dy}{dx} = \square$

2. Split up fractions.

$y = \frac{x^2+3}{\sqrt{x}} = \frac{x^2+3}{x^{\frac{1}{2}}} = \square$

$\rightarrow \frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} - \square$

3. Expand out brackets.

$y = x^2(x-3) = \square$

$\rightarrow \frac{dy}{dx} = \square$

NOT $3x^{-1}$!!!

4. Beware of numbers in denominators!

$y = \frac{1}{3x} = \square \rightarrow \frac{dy}{dx} = \square$

Differentiate the following.

6 $f(x) = \frac{12}{p\sqrt{x}} + x$, where p is a real constant and $x > 0$. Given that $f'(2) = 3$, find p, giving your answer in the form $a\sqrt{2}$ where a is a rational number.

7 $f(x) = (2 - x)^9$.

a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $f(x)$, giving each term in its simplest form.

b If x is small, so that x^2 and higher powers can be ignored, show that $f'(x) \approx 9216x - 2304$.

[MAT 2013 1E]

The expression $\frac{d^2}{dx^2}\left[(2x-1)^4(1-x)^5\right] + \frac{d}{dx}\left[(2x+1)^4(3x^2-2)^2\right]$ is a polynomial of degree:

A) 9
B) 8
C) 7
D) less than 7

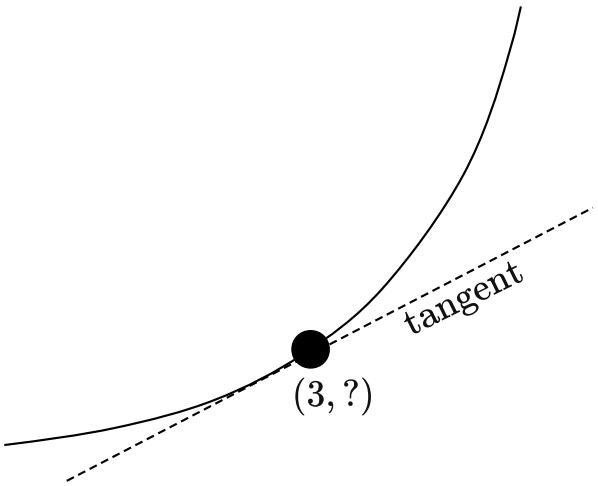
This just means "differentiate twice". The 'second derivative' comes later in this chapter.

Full expansion is not needed. The highest power term in the first

$\mathbf{16x^4 \times (-x)^5 = -16x^9}$

Find the equation of the **tangent** to the curve $y = x^2$ when $x = 3$.

We want to use $y - y_1 = m\,(x - x_1)$ for the tangent (as it is a straight line!).
Therefore we need:
(a) A point (x_1, y_1)
(b) The gradient m.



Gradient function:

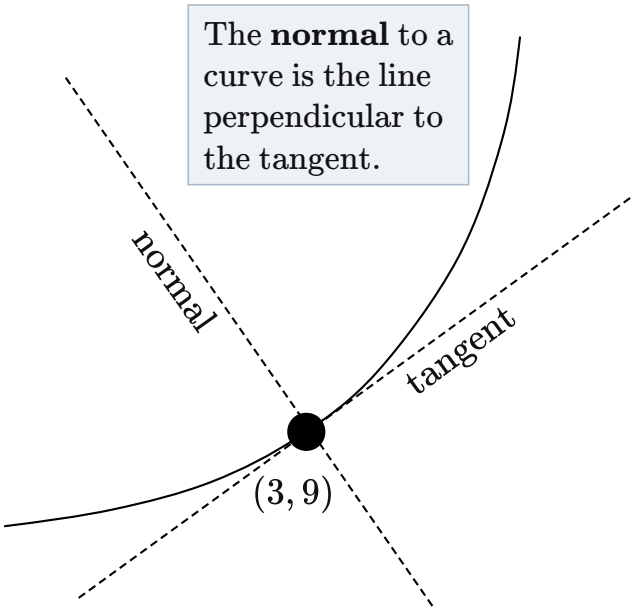
Gradient when $x = 3$: $\mathbf{m=6}$

y-value when $x = 3$: $\mathbf{y=9}$

So equation of tangent:

$$\mathbf{y-9=6\,(x-3)}$$

Find the equation of the **normal** to the curve $y = x^2$ when $x = 3$.



The **normal** to a curve is the line perpendicular to the tangent.

Equation of tangent (from earlier):

$$\mathbf{y-9=6(x-3)}$$

Therefore equation of normal:

$$\mathbf{y-9=}$$

Tip: A **very common error** is for students to accidentally forget whether the question is asking for the tangent or for the normal.

Find the equation of the **normal** to the curve $y = x + 3\sqrt{x}$ when $x = 9$.

When $x = 9$, $y =$

$x + 3x^{\frac{1}{2}}$

$1 + \frac{3}{2}x^{-\frac{1}{2}}$

$8 = -\frac{2}{3}(x$

Tip: I like to use m_T and m_N to make clear to the examiner (and myself) what gradient I've found.

3 Find the coordinates of the point where the tangent to the curve $y = x^2 + 1$ at the point $(2, 5)$ meets the normal to the same curve at the point $(1, 2)$.

4 Find the equations of the normals to the curve $y = x + x^3$ at the points $(0, 0)$ and $(1, 2)$, and find the coordinates of the point where these normals meet.

5 For $\mathrm{f}(x) = 12 - 4x + 2x^2$, find the equations of the tangent and the normal at the point where $x = -1$ on the curve with equation $y = \mathrm{f}(x)$.

6 The point P with x-coordinate $\frac{1}{2}$ lies on the curve with equation $y = 2x^2$. The normal to the curve at P intersects the curve at points P and Q. Find the coordinates of Q. (6 marks)

The 'difference of two cubes':
$a^3 - b^3 = (a-b)\left(a^2 + ab + b^2\right)$
will help for both of these.

1 *[STEP I 2005 Q2]*
The point P has coordinates $\left(p^2, 2p\right)$ and the point Q has coordinates $\left(q^2, 2q\right)$, where p and q are non-zero and $p \neq q$. The curve C is given by $y^2 = 4x$. The point R is the intersection of the tangent to C at P and the tangent to C at Q. Show that R has coordinates $(pq, p+q)$.

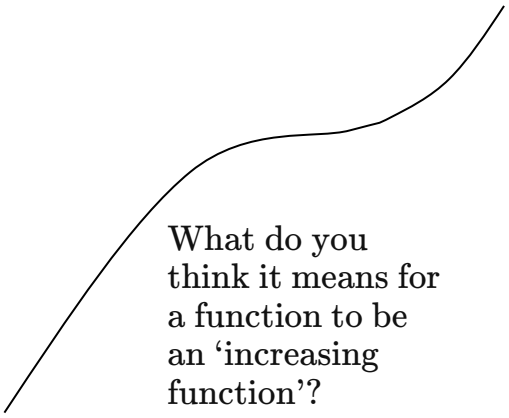
The point S is the intersection of the normal to C at P and the normal to C at Q. If p and q are such that $(1,0)$ lies on the line PQ, show that S has coordinates $\left(p^2+q^2+1, p+q\right)$, and that the quadrilateral $PSQR$ is a rectangle.

Solutions on next slide.

2 *[STEP I 2012 Q4]*
The curve C has equation $xy = \frac{1}{2}$.
The tangents to C at the distinct points $P\left(p, \frac{1}{2p}\right)$ and $Q\left(q, \frac{1}{2q}\right)$, where p and q are positive, intersect at T and the normal to C at these points intersect at N. Show that T is the point $\left(\frac{2pq}{p+q}, \frac{1}{p+q}\right)$
In the case $pq = \frac{1}{2}$, find the coordinates of N. Show (in this case) that T and N lie on the line $y = x$ and are such that the product of their distances from the origin is constant.

This is using something called 'implicit differentiation' (Year 2), but you could easily do:
$y^2 = 4x \to y = 2x^{\frac{1}{2}}$
$\frac{dy}{dx} = x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}} = \frac{2}{y}$

$y^2 = 4x \Rightarrow 2yy' = 4 \Rightarrow y' = \frac{2}{y}$.
Equation of tangent at P: $y - 2p = \frac{1}{p}(x - p^2) \Rightarrow py$
Equation of tangent at Q: $qy = x + q^2$.
Intersect where $qy - q^2 = py - p^2$.
$\Rightarrow (q-p)y = q^2 - p^2$.
$\Rightarrow y = p + q \Rightarrow x = pq$ by substitution. Hence $R(pq$
Equation of normal at P: $y - 2p = -p(x - p^2) \Rightarrow y$
Equation of normal at Q: $y + qx = 2q + q^3$.
Intersect where $2p + p^3 - px = 2q + q^3 - qx$.
$\Rightarrow x(p-q) = 2(p-q) + (p-q)(p^2 + pq + q^2)$ using the identity $p^3 - q^3 = (p-q)(p^2 + pq + q^2)$.
$\Rightarrow x = p^2 + pq + q^2 + 2 \Rightarrow y = -pq(p+q)$ by substitution.
But $(1,0)$ lies on PQ so the gradient of the line segment from P to $(1,0)$ equals the gradient of the line segment from Q to $(1,0)$.
$\Rightarrow \frac{2p}{p^2-1} = \frac{2q}{q^2-1}$.
$\Rightarrow 2pq^2 - 2p = 2qp^2 - 2q \Rightarrow pq(q-p) = p - q \Rightarrow pq = -1$.
$\Rightarrow S$, where the normals intersect, has coordinates $(p^2 + q^2 + 1, p + q)$.
Obviously, SP is perpendicular to PR and QS is perpendicular to QR, because each of these is a tangent-normal pair.
Furthermore, the gradient of $PR\times$ the gradient of $QR = \frac{1}{p} \times \frac{1}{q} = \frac{1}{pq} = -1$, so PR is perpendicular to QR.
Also, the gradient of $PS\times$ the gradient of $QS = -p \times -q = pq = -1$, so PS is perpendicular to QS.
Therefore all four angles are right angles, proving that $PSQR$ is a rectangle. It is not sufficient to consider only the lengths of the sides, since the quadrilateral could be a parallelogram.

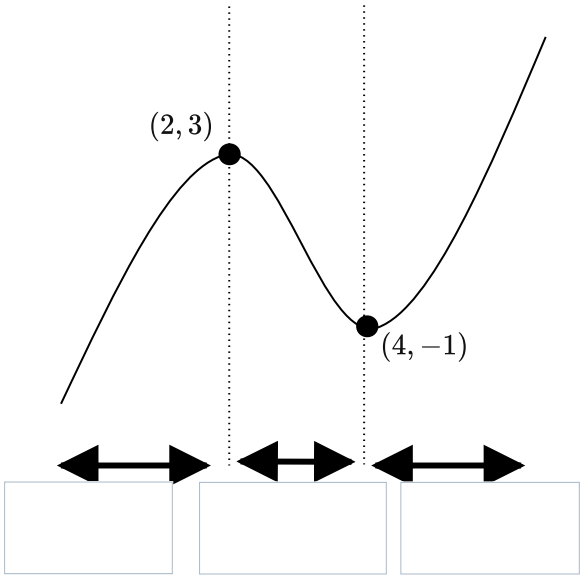


What do you think it means for a function to be an 'increasing function'?

An increasing function is one whose gradient is always at least 0.
$f'(x) \geq 0$ for all x.

It would be '**strictly increasing**' if $f(x) > 0$ for all x, i.e. is not allowed to go horizontal.

A function can also be increasing and decreasing in certain intervals.



We could also write "$f(x)$ is decreasing in the interval $[2, 4]$"
[a,b] represents all the real numbers between a and b inclusive, i.e:

$$[a, b] = \{x : a \leq x \leq b\}$$

Show that the function $f(x) = x^3 + 6x^2 + 21x + 2$ is increasing for all real values of x.

Find the interval on which the function $f(x) = x^3 + 3x^2 - 9x$ is decreasing.

Tip: To show a quadratic is always positive, complete the square, then indicate the squared term is always at least 0.

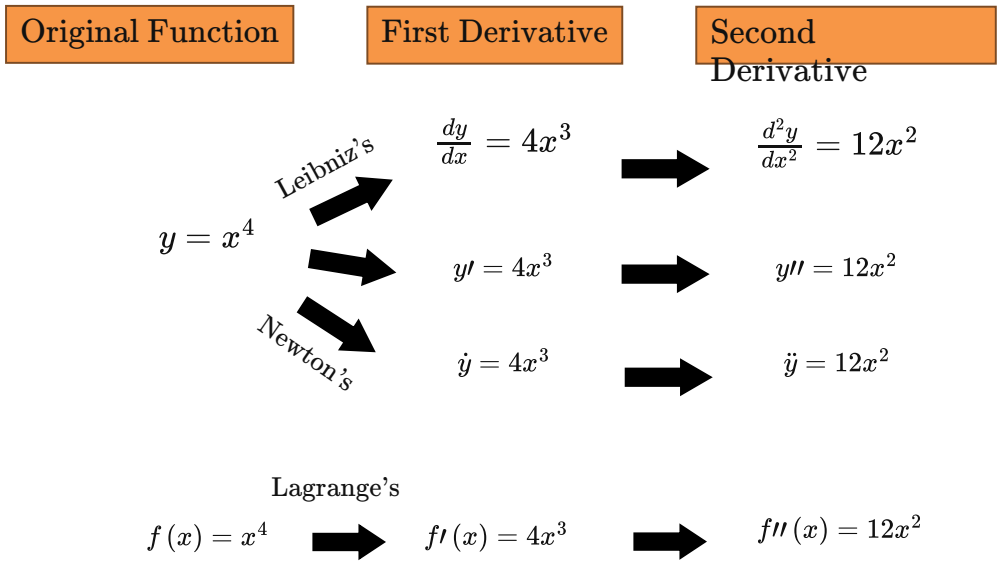
Show that the function $f(x) = x^3 + 16x - 2$ is increasing for all real values of x.

Find the interval on which the function $f(x) = x^3 + 6x^2 - 135x$ is decreasing.

3 Show that the function $f(x) = 4 - x(2x^2 + 3)$ is decreasing for all $x \in \mathbb{R}$.

4 a Given that the function $f(x) = x^2 + px$ is increasing on the interval $[-1, 1]$, find one possible value for p.
b State with justification whether this is the only possible value for p.

When you differentiate once, the expression you get is known as the **first derivative**. Unsurprisingly, when we differentiate a second time, the resulting expression is known as the **second derivative**. And so on...



You can similarly have the third derivative ($\frac{d^3y}{dx^3}$), although this is no longer in the A Level syllabus. We'll see why might use the second derivative soon...

How does the notation $\frac{d^2y}{dx^2}$ work? Why are the squareds where they are?

Suppose that $y = x^3 + 1$.
Then when we write $\frac{dy}{dx}$, we're effectively doing $\frac{d(x^3+1)}{dx}$ (by substitution), although this would typically be written:

$$\frac{d}{dx}\left(x^3 + 1\right)$$

The $\frac{d}{dx}(\dots)$ notation is quite handy, because it behaves as a function and allows us to write the original expression and the derivative within a single equation:

$$\frac{d}{dx}\left(x^3 + 1\right) = 3x^2$$

Therefore, if we wanted to differentiate y twice, we'd do:

$\frac{d}{dx}\left(\frac{d}{dx}(y)\right)$
$= \frac{d^2}{dx^2}(y)$
$= \frac{d^2y}{dx^2}$

If $y = 3x^5 + \frac{4}{x^2}$, find $\frac{d^2y}{dx^2}$.

If $f(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}$, find $f''(x)$.

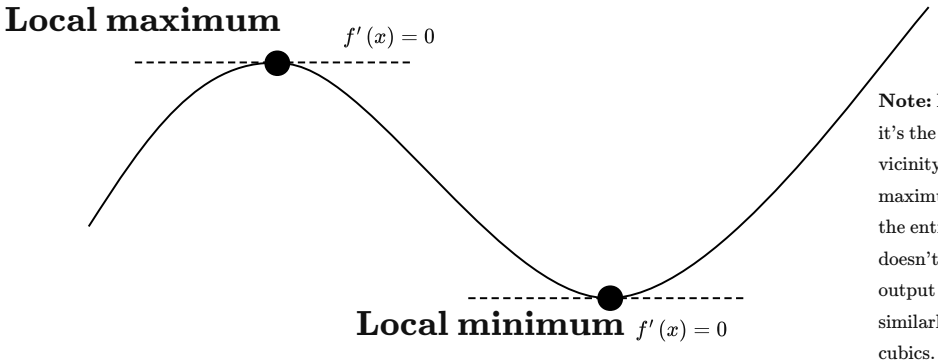
If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find $\frac{d^2y}{dx^2}$.

3 Given that $y = (2x - 3)^3$, find the value of x when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$.

4 $\mathrm{f}(x) = px^3 - 3px^2 + x^2 - 4$. When $x = 2$, $\mathrm{f}''(x) = -1$. Find the value of p.

Stationary/Turning Points 48

A stationary point is where the gradient is 0, i.e. $f'(x) = 0$.



Note: It's called a '**local**' maximum because it's the function's largest output within the vicinity. Functions may also have a '**global**' maximum, i.e. the maximum output across the entire function. This particular function doesn't have a global maximum because the output keeps increasing up to infinity. It similarly has no global minimum, as with all cubics.

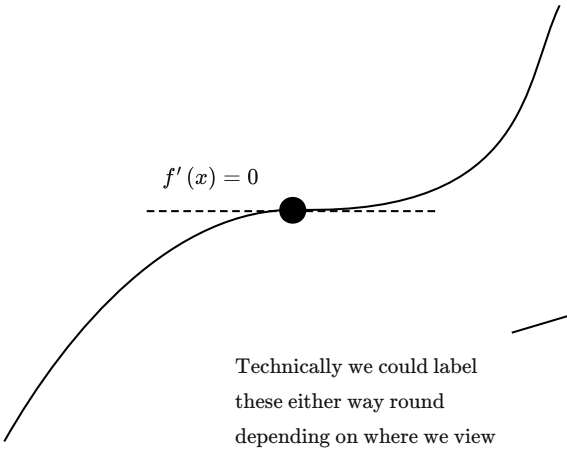
Find the coordinates of the turning points of $y = x^3 + 6x^2 - 135x$

Find the least value of
$f(x) = x^2 - 4x + 9$

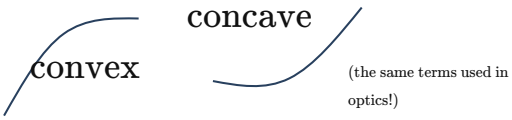
Find the turning point of
$y = \sqrt{x} - x$

Note: Method 2 is only applicable for quadratic functions. For others, differentiation must be used.

There's a third type of stationary point (that we've encountered previously):



A point of inflection is where the curve changes from convex ⟶ concave (or vice versa).

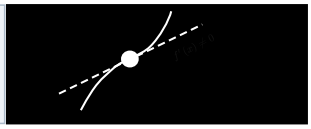


(the same terms used in optics!)

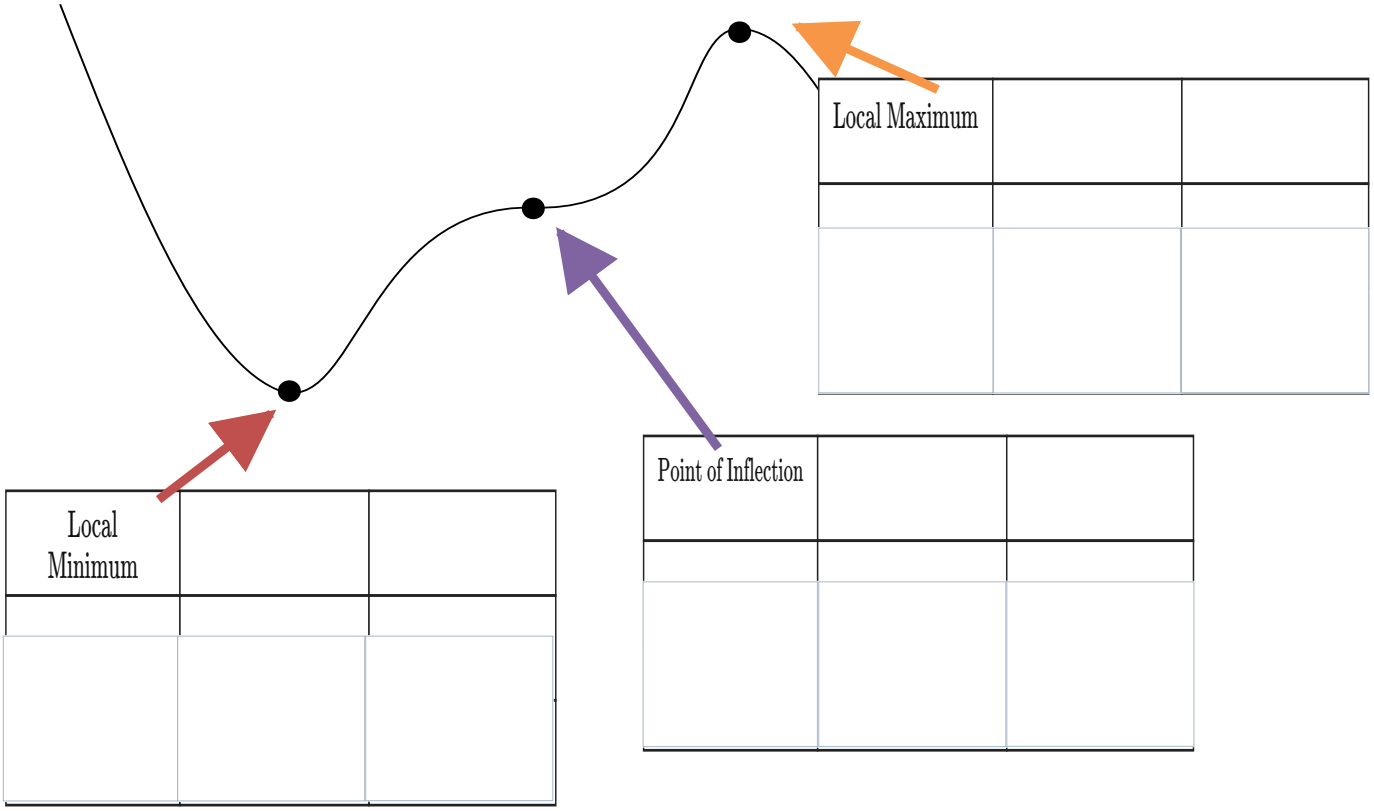
Technically we could label these either way round depending on where we view the curve from. What's important is that the **concavity changes**.

i.e. the line curves in one direction before the point of inflection, then curves in the other direction after.

Side Note: Not all points of inflection are stationary points, as can be seen in the example on the right.
A point of inflection which **is** a stationary point is known as a *saddle point.*



Method 1: Look at gradient just before and just after point.



Method 1: Look at gradient just before and just after point.

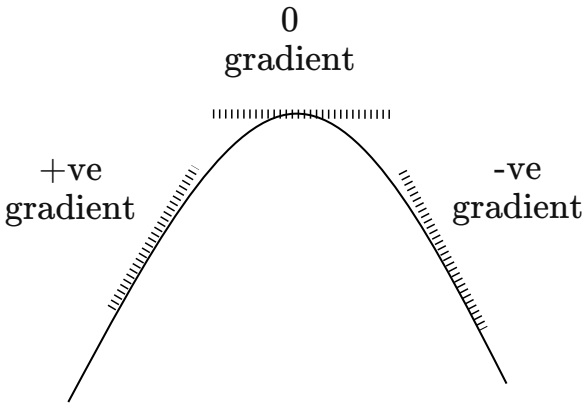
Find the stationary point on the curve with equation $y = x^4 - 32x$, and determine whether it is a local maximum, a local minimum or a point of inflection.

The method of substituting values of x just before and after is a bit cumbersome. It also has the potential for problems: what if two different types of stationary points are really close together?

Recall the gradient gives a measure of the **rate of change** of y, i.e. how much the y value changes as x changes.

Thus by differentiating the gradient function, the **second derivative tells us the rate at what the gradient is changing**.

Thus if the second derivative is positive, the gradient is increasing.
If the second derivative is negative, the gradient is decreasing.



At a maximum point, we can see that as x increases, the gradient is decreasing from a positive value to a negative value.

$$\therefore \frac{d^2y}{dx^2} < 0$$

✎ At a stationary point $x = a$:
- If $f''(a) > 0$ the point is a local minimum.
- If $f''(a) < 0$ the point is a local maximum.
- If $f''(a) = 0$ it could be any type of point, so resort to Method 1.

The stationary point of $y = x^4 - 32x$ is $(2, -48)$.
Use the second derivative to classify this stationary point.

I will eventually do a 'Just for your Interest…' thingy on why we can't classify the point when $f''(x) = 0$, and how we could use the **third derivative**!

The curve with equation $y = x^2 - 32\sqrt{x} + 20$, $x > 0$, has a stationary point P. Use calculus
(a) to find the coordinates of P, (6)
(b) to determine the nature of the stationary point P. (3)

All the way back in Chapter 4, we used features such as intercepts with the axes, and behaviour when $x \to \infty$ and $x \to -\infty$ in order to sketch graphs. Now we can also find stationary/turning points!

[Textbook] By first finding the stationary points, sketch the graph of $y = \frac{1}{x} + 27x^3$

5 By considering the gradient on either side of the stationary point on the curve $y = x^3 - 3x^2 + 3x$, show that this point is a point of inflection. Sketch the curve $y = x^3 - 3x^2 + 3x$.

6 Find the maximum value and hence the range of values for the function $\mathrm{f}(x) = 27 - 2x^4$.

7 $f(x) = x^4 + 3x^3 - 5x^2 - 3x + 1$.

a Find the coordinates of the stationary points of $f(x)$, and determine the nature of each.

b Sketch the graph of $y = f(x)$.

1 [MAT 2014 1C] The cubic

$$y = kx^3 - (k+1)x^2 + (2-k)x - k$$

has a turning point, that is a minimum, when $x = 1$ precisely for

A) $k > 0$
B) $0 < k < 1$
C) $k > \frac{1}{2}$
D) $k < 3$
E) all values of k

2 *[MAT 2004 1B]* The smallest value of the function:

$$f(x) = 2x^3 - 9x^2 + 12x + 3$$

In the range $0 \leq x \leq 2$ is what?

3 *[MAT 2001 1E]* The maximum gradient of the curve $y = x^4 - 4x^3 + 4x^2 + 2$ in the range $0 \leq x \leq 2\frac{1}{5}$ occurs when:

A) $x = 0$
B) $x = 1 - \frac{1}{\sqrt{3}}$
C) $x = 1 + \frac{1}{\sqrt{3}}$
D) $x = 2\frac{1}{5}$

Hint: When two curves touch, their y values must match, but what else must also match?

4 [STEP I 2007 Q8] A curve is

$$y = ax^3 - 6ax^2 + (12$$

where a is a real number. Show that this curve touches the curve with equation $y = x^3$ at $(2, 8)$. Determine the coordinates of any other point of intersection of the two curves.

(i) Sketch on the same axes these two curves when $a = 2$.

(ii) ... when $a = 1$ (iii) when $a = -2$

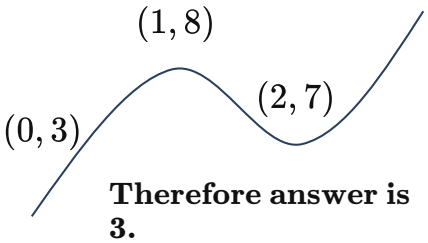
1 [MAT 2014 1C] The cubic
$$y = kx^3 - (k+1)x^2 + (2-k)x - k$$
has a turning point, that is a minimum, when $x = 1$ precisely for

A) $k > 0$
B) $0 < k < 1$
C) $k > \frac{1}{2}$
D) $k < 3$
E) all values of k

$$\mathbf{\frac{dy}{dx} = 3kx^2 - 2(k+1)x + (2-k)}$$
When x=1, $\frac{dy}{dx} = 3k - 2k - 2 + 2 - k \equiv 0$
Therefore there is a turning point for all values of k. However, this must be a minimum.
$$\mathbf{\frac{d^2y}{dx^2} = 6kx - 2(k+1)}$$
When x=1, $\frac{d^2y}{dx^2} = 4k - 2$
If minimum: $4k - 2 > 0 \quad \rightarrow \quad k > \frac{1}{2}$

2 *[MAT 2004 1B]* The smallest value of the function:
$$f(x) = 2x^3 - 9x^2 + 12x + 3$$
In the range $0 \leq x \leq 2$ is what?

$f'(x) = 6x^2 - 18x + 12 = 0$
$x^2 - 3x + 2 = 0$
$(x-1)(x-2) = 0$
$x = 1 \quad \rightarrow \quad f(1) = 8$
$x = 2 \quad \rightarrow \quad f(2) = 7$
At start of range: $f(0) = 3$



Therefore answer is 3.

3 [MAT 2001 1E] The maximum gradient of the curve $y = x^4 - 4x^3 + 4x^2 + 2$ in the range $0 \leq x \leq 2\frac{1}{5}$ occurs when:

A) $x = 0$
B) $x = 1 - \frac{1}{\sqrt{3}}$
C) $x = 1 + \frac{1}{\sqrt{3}}$
D) $x = 2\frac{1}{5}$

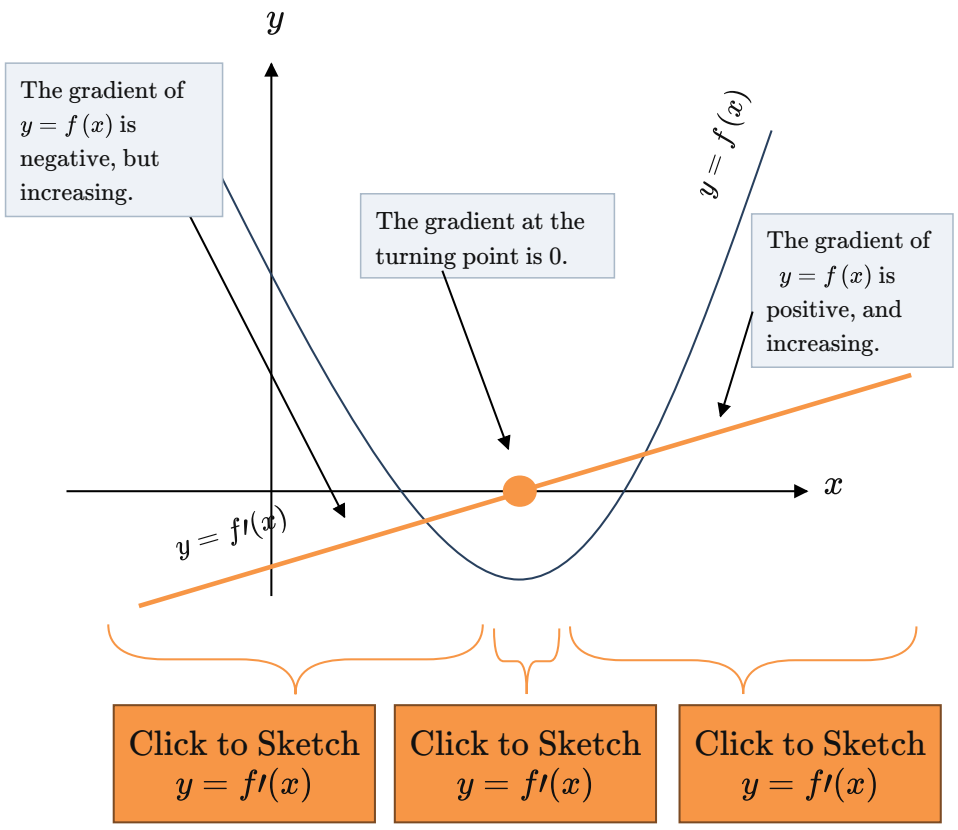
Gradient: $\frac{dy}{dx} = 4x^3 - 12x^2 + 8x$
So find max value of $4x^3 - 12x^2 + 8x$
$\frac{d^2y}{dx^2} = 12x^2 - 24x + 8 = 0$
$3x^2 - 6x + 2 = 0$
$x = 1 \pm \frac{1}{\sqrt{3}}$
Due to the shape of a cubic, we have the a local maximum gradient at $x = 1 - \frac{1}{\sqrt{3}}$ and a local minimum at $x = 1 + \frac{1}{\sqrt{3}}$
In range $0 \leq x \leq 2\frac{1}{5}$, answer must either be $x = 1 + \frac{1}{\sqrt{3}}$ or $x = 2\frac{1}{5}$. Substituting these into y yields a higher value for the latter, so answer is (D).

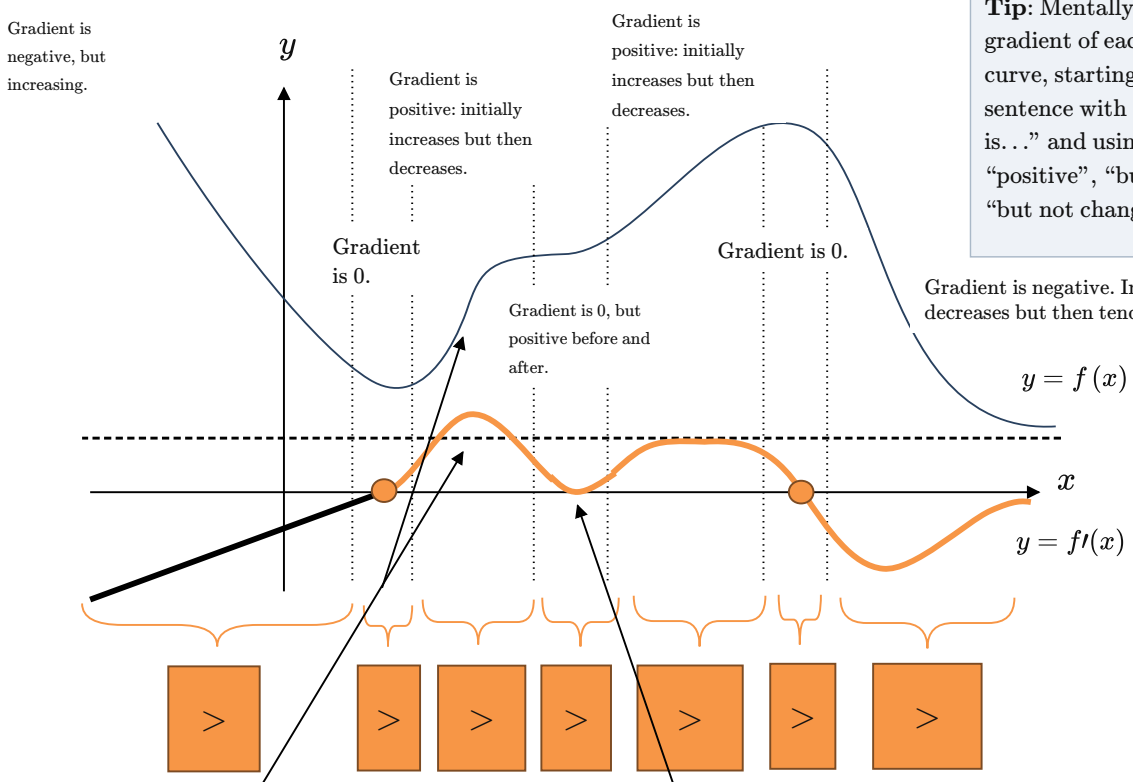
4 [STEP I 2007 Q8] A curve is given by:
$y = ax^3 - 6ax^2 + (12a + 12)x - (8a + 16)$
where a is a real number. Show that this curve touches the curve with equation $y = x^3$ at $(2, 8)$. Determine the coordinates of any other point of intersection of the two curves.
(i) Sketch on the same axes these two curves when $a = 2$.
(ii) ... when $a = 1$
(iii) ... when $a = -2$

Consider $f(x) = ax^3 - 6ax^2 + (12a + 12)x - (8a + 16)$.
Since $f(2) = 8$ and $f'(2) = 12$, the curve $y = f(x)$ touches $y = x^3$ at $(2, 8)$: notice that both calculations are necessary to prove that the curves touch.
To find the other intersection point, let $f(x) = x^3$.
$\Rightarrow (a - 1)x^3 - 6ax^2 + (12a + 12)x - (8a + 16) = 0$.
Substituting $x = 2$: $8(a - 1) - 24a + 2(12a + 12) - (8a + 16) = 0$.
$\Rightarrow (x - 2)[(a - 1)x^2 - (4a + 2)x + (4a + 8)] = 0$ (notice that factorising by inspection is much easier than using a method such as long division).
Substituting $x = 2$ into the quadratic factor:
$4(a - 1) - 2(4a + 2) + (4a + 8) = 0$.
$\Rightarrow (x - 2)(x - 2)[(a - 1)x - (2a + 4)] = 0$.
So the other intersection point has coordinates $\left(\frac{2a+4}{a-1}, \left[\frac{2a+4}{a-1}\right]^3\right)$.
(i) When $a = 2$, $\frac{2a+4}{a-1} = 8$.
Hence the two graphs touch at $(2, 8)$ and intersect at $(8, 512)$.
$y = 2x^3 - 12x^2 + 36x - 32$ has no turning points: consider the derivative $6x^2 - 24x + 36 = 0$.
(ii) When $a = 1$, $\frac{2a+4}{a-1}$ is undefined.
Hence the two graphs touch at $(2, 8)$ and do not intersect elsewhere.
$y = x^3 - 6x^2 + 24x - 24$ has no turning points: consider the derivative $3x^2 - 12x + 24 = 0$.
(iii) When $a = -2$, $\frac{2a+4}{a-1} = 0$.
Hence the two graphs touch at $(2, 8)$ and intersect at $(0, 0)$.
$y = -2x^3 + 12x^2 - 12x$ turns when $x = 2 \pm \sqrt{2} \approx 3.4$ and 0.6: consider the derivative $-6x^2 + 24x - 12 = 0$.

The new A Level specification specifically mentions being able to sketch $y = f\prime(x)$.

If you know the function $f\prime(x)$ explicitly (e.g. because you differentiated $y = f(x)$), you can use your knowledge of sketching straight line/ quadratic/cubic graphs. But in other cases **you won't be given the function explicitly**, but **just the sketch**.

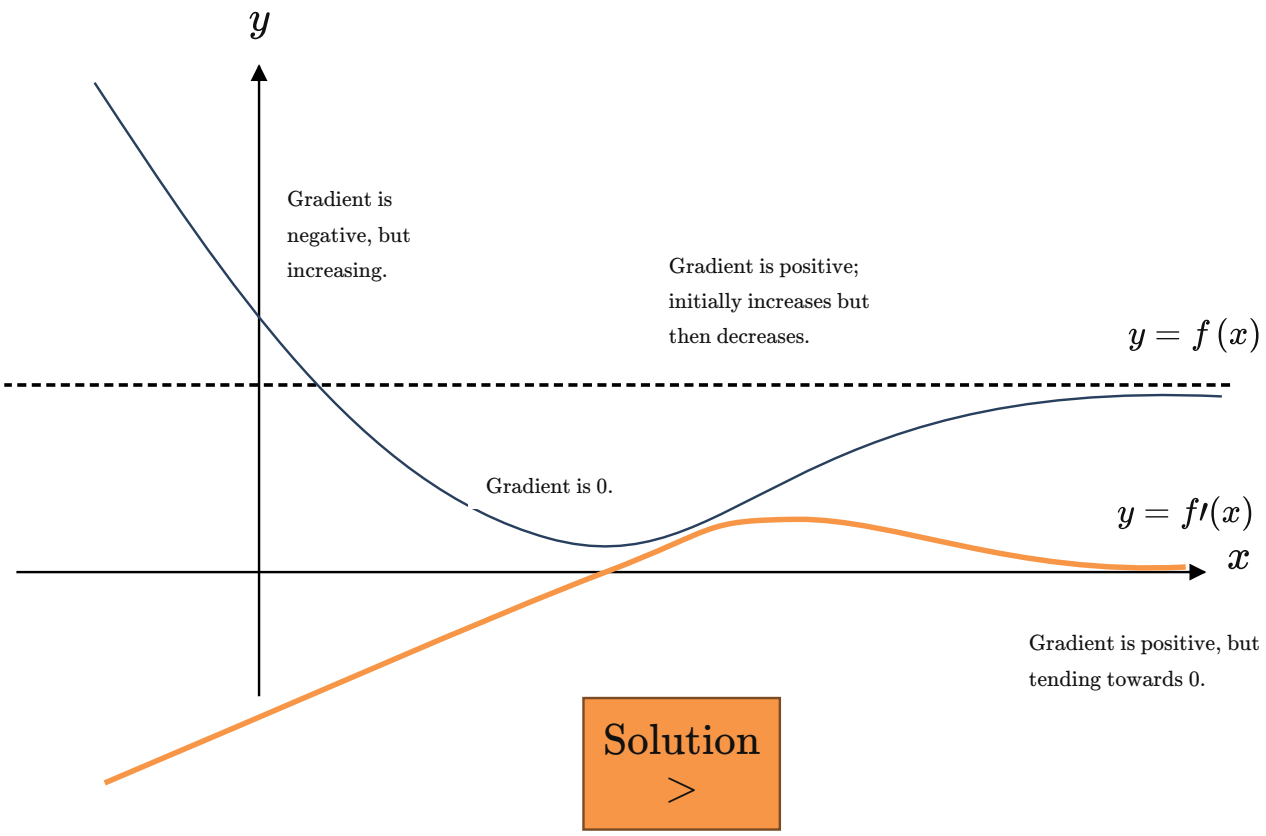




Tip: Mentally describe the gradient of each section of the curve, starting your mental sentence with "The gradient is. . ." and using terms like "positive", "but increasing", "but not changing", etc.

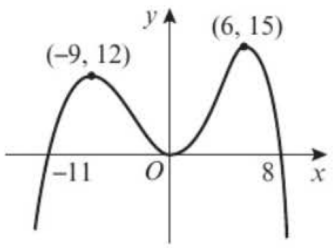
Tip: The gradient is momentarily not changing so the gradient of the gradient is 0. We get a **turning point** in $y = f'(x)$ whenever the **curve has a point of inflection**.

Again, on $y = f(x)$ this is a point of inflection so the gradient function has a turning point. It is also a stationary point, so $f'(x) = 0$. i.e. The $f\prime(x)$ curve touches the x-axis.

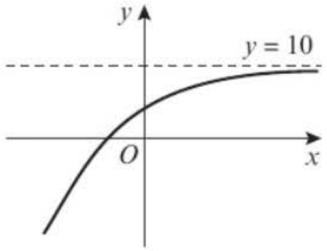


1 For each graph given, sketch the graph of the corresponding gradient function on a separate set of axes. Show the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.

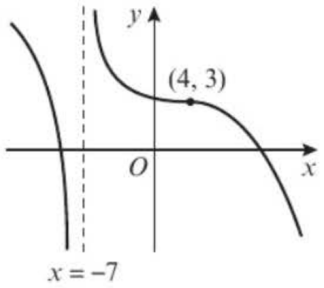
a



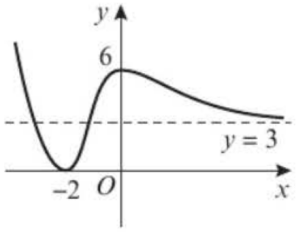
b



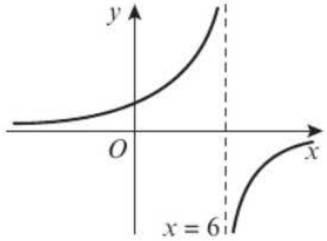
c



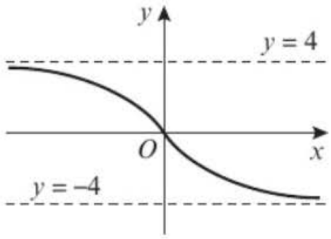
d



e



f



Solutions here

2 $f(x) = (x + 1)(x - 4)^2$.

a Sketch the graph of $y = f(x)$.

b On a separate set of axes, sketch the graph of $y = f'(x)$.

c Show that $f'(x) = (x - 4)(3x - 2)$.

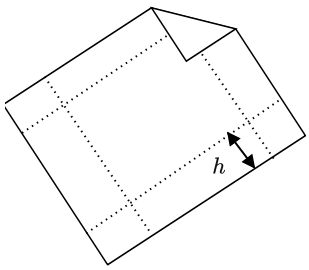
d Use the derivative to determine the exact coordinates of the points where the gradient function cuts the coordinate axes.

1 [MAT 2015 1B]

$$f(x) = (x+a)^n$$

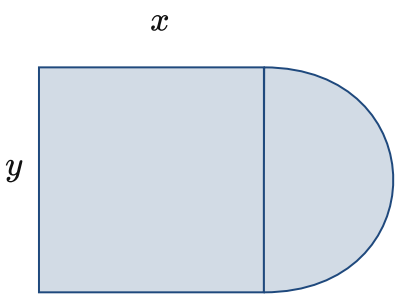
where a is a real number and n is a positive whole number, and $n \geq 2$. If $y = f(x)$ and $y = f\prime(x)$ are plotted on the same axes, the number of intersections between $f(x)$ and $f'(x)$ will:

A) always be odd
B) always be even
C) depend on a but not n
D) depend on n but not a
E) depend on both a and n.



We have a sheet of A4 paper, which we want to fold into a cuboid. What height should we choose for the cuboid in order to maximise the volume?

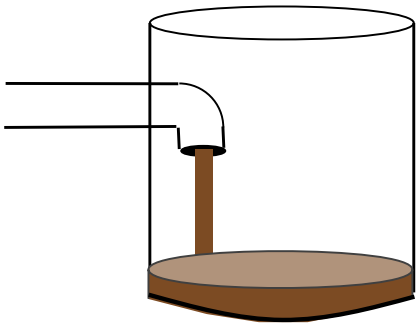
These are examples of **optimisation problems**: we're trying to maximise/minimise some quantity by choosing an appropriate value of a variable that we can control.



We have 50m of fencing, and want to make a bear pen of the following shape, such that the area is maximised. What should we choose x and y to be?

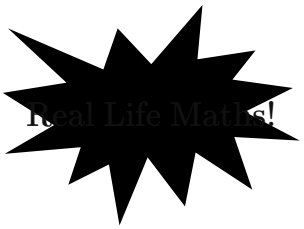
Up to now we’ve had y in terms of x, where $\frac{dy}{dx}$ means “the **rate** at which y changes with respect to x”.

But we can use similar gradient function notation for other **physical quantities**.



A sewage container fills at a rate of 20 cm^3 per second.

How could we use appropriate notation to represent this?

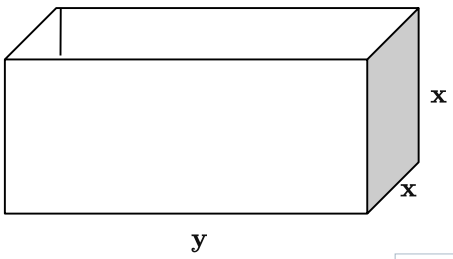


$$\frac{dV}{dt} = 20cm^3/s$$

“The rate at which the volume V changes with respect to time t.”

Optimisation problems in an exam usually follow the following pattern:

- There are 2 variables involved (you may have to introduce one yourself), typically lengths.
- There are expressions for **two different physical quantities**:
  - One is a **constraint**, e.g. “the surface area is 20cm^2”.
  - The **other we wish to maximise/minimise**, e.g. “we wish to maximise the volume”.
- We use the constraint to **eliminate one of the variables** in the latter equation, so that it is then **just in terms of one variable**, and we can then use differentiation to find the turning point.



[Textbook] A large tank in the shape of a cuboid is to be made from 54m^2 of sheet metal. The tank has a horizontal base and no top. The height of the tank is x metres. Two of the opposite vertical faces are squares.

a) Show that the volume, V m^3, of the tank is given by $V = 18x - \frac{2}{3}x^3$.

We need to introduce a second variable ourselves so that we can find expressions for the surface area and volume.

These are the two equations mentioned in the guidance: one for surface area and one volume.

b) Given that x can vary, use differentiation to find the maximum or minimum value of V.

Once we have the ‘optimal’ value of x, we sub it back into V to get the best possible volume.

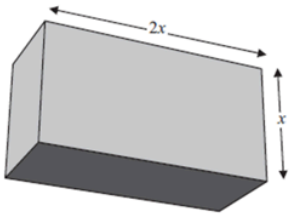


Figure 2

A cuboid has a rectangular cross-section where the length of the rectangle is equal to twice its width, x cm, as shown in Figure 2.

The volume of the cuboid is 81 cubic centimetres.

(a) Show that the total length, L cm, of the twelve edges of the cuboid is given by
$L = 12x + \frac{162}{x^2}$.

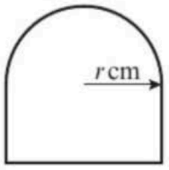
6 A rectangular garden is fenced on three sides, and the house forms the fourth side of the rectangle.

a Given that the total length of the fence is 80 m, show that the area, A, of the garden is given by the formula $A = y(80 - 2y)$, where y is the distance from the house to the end of the garden.

b Given that the area is a maximum for this length of fence, find the dimensions of the enclosed garden, and the area which is enclosed.

7 A closed cylinder has total surface area equal to 600π.
a Show that the volume, $V\,\text{cm}^3$, of this cylinder is given by the formula $V = 300\pi r - \pi r^3$, where $r\,\text{cm}$ is the radius of the cylinder.
b Find the maximum volume of such a cylinder.

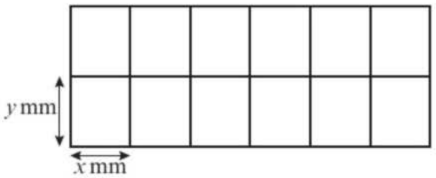
9 A shape consists of a rectangular base with a semicircular top, as shown.



a Given that the perimeter of the shape is $40\,\text{cm}$, show that its area, $A\,\text{cm}^2$, is given by the formula
$A = 40r - 2r^2 - \frac{\pi r^2}{2}$

where $r\,\text{cm}$ is the radius of the semicircle. (2 marks)

10 The shape shown is a wire frame in the form of a large rectangle split by parallel lengths of wire into 12 smaller equal-sized rectangles.



a Given that the total length of wire used to complete the whole frame is 1512 mm, show that the area of the whole shape, A mm^2, is given by the formula
$A = 1296x - \frac{108x^2}{7}$

where x mm is the width of one of the smaller rectangles. (4 marks)

1 [STEP I 2006 Q2] A small goat is tethered by a rope to a point at ground level on a side of a square barn which stands in a large horizontal field of grass. The sides of the barn are of length $2a$ and the rope is of length $4a$. Let A be the area of the grass that the goat can graze.
Prove that $A \leq 14\pi a^2$ and determine the minimum value of A.